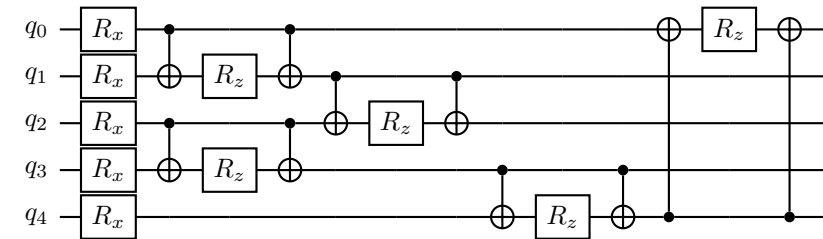
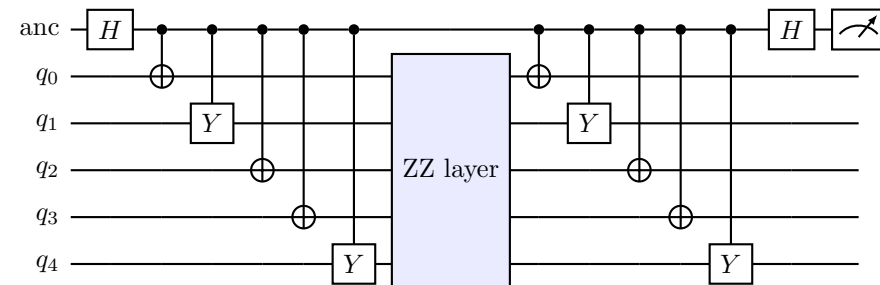
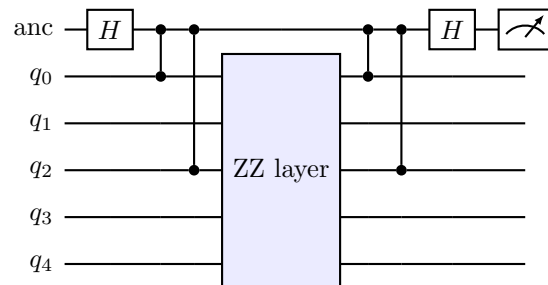
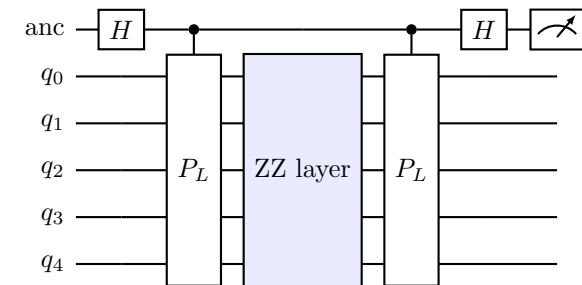


(a) **One Trotter step of the payload** ($n = 5$ ring; bonds ordered even, odd, closing; $\exp(-i\frac{\phi}{2}Z_jZ_k) = CX_{jk}R_z^{(k)}(\phi)CX_{jk}$; 10 noisy CX per step)



(b) **Constrained twirled check (Stage 3, 3b)**: P_L drawn per shot, $\frac{1}{2}$ a Z-string on a random subset, $\frac{1}{2}$ a string with X or Y on every qubit; $P_R = P_L$; no repair.

Left: generic form. Middle: $P_L = Z_0Z_2$ (2+2 controlled-Paulis). Right: $P_L = X_0Y_1X_2X_3Y_4$ (5+5). Average over the twirl: 7.5 noisy gadget gates per step.



(c) **Repaired full twirl (Stage 3c)**: P_L uniform over all 4^5 Paulis; for every bond (j, k) with $\{P_L, Z_jZ_k\} = 0$ a controlled- X_k from the same ancilla is inserted immediately before and after $R_z^{(k)}$ (red). Example $P_L = X_0$: anticommutes with bonds $(0, 1)$ and $(4, 0)$, so two repaired rotations (R_z on q_1 and on q_0); $P_R = P_L$.

Average over the twirl: 7.5 controlled-Paulis + 5.0 controlled- X = 12.5 noisy gadget gates per step.

